Fiscal Year:	FY 2006	**Task Last Updated:**	FY 03/31/2006
PI Name:	Peterson, Kevin Ph.D.		
Project Title:	Using Molecular Clocks And Regulatory Networks To Test Origin-Of-Bilateria Scenarios		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Evolutionary biology		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Musculoskeletal Biology		
Space Biology Special Category:	None		
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Comments:			
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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	Because of the wealth of both phylogenetic and developmental data, there is now a tremendous interest in the origin of bilaterian metazoans and the associated paleontological events such as the Cambrian explosion. It is now generally accepted that the latest common ancestor of bilaterians was a fairly complex animal complete with anterior-posterior and dorsal-ventral axes, and a central nervous system. Nonetheless, it is still unclear if the origin of this latest common ancestor shortly preceded the Cambrian explosion, as suggested by the fossil record, or if there was a significant but cryptic period of bilaterian evolution before their first appearance in the fossil record, as suggested by most molecular-clock studies. Understanding which stage of the bilaterian life cycle is primitive for Bilateria, the larval stage or the adult stage, is crucial for deciphering the temporal incongruence between the fossil record and these molecular-clock analyses. To test for homology of larval types, we cloned a gene in the purple sea urchin		

Task Description:	Strongylocentrotus purpuratus called NK2.1, a member of the NK-2 class of homeodomain transcription factors. NK2.1 is expressed at the apical pole of the sea urchin, and transcripts are eventually restricted to the larval apical organ. Because the apical organ is restricted to primary larval forms, homology of the apical organ implies homology of the larva. In order to test this homology, we want to test for the conservation of gene batteries underlying apical pole development in sea urchins and in gastropod molluscs. We have already identified several important upstream factors in the sea urchin, and are in the process of examining these same genes in the mollusc. Determining the polarity of the larval vs. adult bilaterian body plan will then allow for a better understanding of major events in the history of life on Earth including the Cambrian explosion. Our primary objetive was completed this past year with the elucidiation of the regulatory network underlying apical organ development in the sea urchin, and examination of this network in the gastropod. Not unexpectedly, this network is not wired the same in the gastropod suggesting that the two different larval forms arose convergently, and the last common ancestor of bilaterians was not constructed like a bilaterian larvae. The significance is that these data are consistenet with zooplankton evolving multiple times indepdenently in the earliest Cambrian, and feeding larvae evolving in the earliest Ordovician, both of which important components of many Cambrian explosion scenarios.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Ascertaining the primitive life cycle of bilaterians will shed tremendous insight into the ecological ramifications of surviving catastrophic environmental perturbations such as "snowball earth". Indeed, current discussions among snowball-earth workers have focused, in part, on the inferred ecology of the animals surviving the glaciation(s) (see Runnegar 2000, and discussion by Schrag & Hoffman 2001 and Hyde et al. 2001). Thus, a proper understanding of the earliest stages of bilaterian evolution is necessary in order to properly understand the geological and paleontological events of the Late Neoproterozoic. Finally, given that bilaterians may have been present for ~ 100 Ma before they made their first appearance in the fossil record, small but potentially complex metazoans may have thrived and diversified without leaving a trace on our own planet in a relatively recent time frame. Discussions centered on discovering complex life throughout the universe should keep this most interesting possibility in mind.
Task Progress:	During the past year we have elucidated the role NK2.1 plays during the development of the apical organ in the sea urchin S. purpuratus. We have isolated several downstream targets and all of them are ciliary genes expressed in the cells destined to give rise to the apical tuft. A knockdown of SpNK2.1 results in either the abrogation or over-expression of these genes, and results in an embryo that lacks the apical tuft. We then cloned all of these genes from the gastropod mollusc H. rufescens and although NK2.1 and the ciliary genes are expressed apically, NK2 and the ciliary genes are not expressed in the same cells demonstrating that NK2.1 cannot be a direct upstream regulator like it is in the sea urchin. The importance of this observation is that it is the first evidence that these two larval forms arose independently, and hence the last common ancestor of bilaterians was most likely not a ciliated planktic form but a direct-developing benthic animal. In addition, further molecular clock work has now shown that these larval forms most likely arose convergently in the latest Precambrian, and that feeding larvae evolved multiple times independently. Ecological considerations strongly support the accuracy of this molecular clock, and support the notion that the last common ancestor of bivalves and molluscs, which the above-mentioned data suggest did not have a larval form, evolved sometime around 580 million years ago.
Bibliography Type:	Description: (Last Updated: 02/23/2007)
Articles in Peer-reviewed Journals	Peterson KJ, Butterfield NJ. "Origin of the Eumetazoa – testing ecological predictions of molecular clocks against the Proterozoic fossil record." Proc Natl Acad Sci U S A. 2005 Jul 5;102(27):9547-52. PMID: 15983372 , Jul-2005
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Journal/Magazine covers	Peterson KJ, Butterfield NJ. "Cover in journal PNAS for article, Origin of the Eumetazoa – testing ecological predictions of molecular clocks against the Proterozoic fossil record." Proc Natl Acad Sci U S A. 2005 Jul 5;102(27):9547-52. PMID: 15983372 , Jul-2005